

Zicheng Wang

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ACADEMIC APPOINTMENTS

The Chinese University of Hong Kong, Shenzhen Assistant Professor, School of Data Science	Shenzhen, Guangdong 2023 - present
University of Minnesota Post-Doctoral Associate, Department of Laboratory Medicine and Pathology	Minneapolis, MN 2021 - 2023

EDUCATION

University of Minnesota Ph.D. in Industrial and Systems Engineering Dissertation Title: “Mathematical Analysis of Cancer Recurrence” Committee: Kevin Leder (advisor), William Cooper, Sherwin Doroudi and Jasmine Foo	Minneapolis, MN 2015 - 2021
University of Minnesota B.A. in Mathematics	Minneapolis, MN 2011 - 2015

RESEARCH INTERESTS

My research interests lie in the interface between theories in **Stochastic Processes** and **Cancer Data Science** (and **Operations Management**). My research aims to analyze the evolution of **Stochastic Systems** using both a **Theoretical** and a **Data Science** approach and solving related **Optimization** problems.

PUBLICATIONS AND PREPRINTS (*AUTHORS IN ALPHABETICAL ORDER)

• Stochastic Processes and Cancer Data Science

*Kevin Leder, Xin Liu, and Zicheng Wang. Splitting Algorithms for Rare Events of Semimartingale Reflecting Brownian Motions. **Stochastic Systems**, 11(4), 291-325 (2021)
Link: <https://pubsonline.informs.org/doi/abs/10.1287/stsy.2021.0076>

*Pranav Hanagal, Kevin Leder, and Zicheng Wang. Large Deviations of Cancer Recurrence Timing. **Stochastic Processes and their Applications**, 147, 1-50 (2022)
Link: <https://www.sciencedirect.com/science/article/pii/S0304414922000011>

Zicheng Wang, Yunong Xia, Lauren Mills, Athanasios N. Nikolakopoulos, Nicole Maeser, Jason Sheltzer, and Ruping Sun. Evolving Copy Number Gains Promote Tumor Expansion and Bolster Mutational Diversification. **Nature Communications**, 15(1), 2025 (2024)
Link: <https://www.nature.com/articles/s41467-024-46414-5>

*Kevin Leder, Ruping Sun, Zicheng Wang, Xuanming Zhang. Parameter Estimation from Single Patient, Single Time-Point Sequencing Data of Recurrent Tumors. **Journal of Mathematical Biology**, 89, 51 (2024).
Link: <https://link.springer.com/article/10.1007/s00285-024-02149-x>

*Kevin Leder and Zicheng Wang. Clonal Diversity at Early Cancer Recurrence. **Bulletin of Mathematical Biology**, Accepted, 2026
Link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5081427

*Kevin Leder, Zicheng Wang, Xuanming Zhang. Parameter Estimation in Recurrent Tumor Evolution with Finite Carrying Capacity. Under Review, 2026
Link: <https://arxiv.org/abs/2510.01078>

Chenyu Wu, Kevin Leder, Zicheng Wang, Ruping Sun. LeafRank: A phylodynamic framework for inferring relative fitness from single-cell phylogenies in chromosomally unstable tumors. Under Review, 2026
Link: <https://www.biorxiv.org/content/10.64898/2026.07.06.736651v2>

- Operations Management

*Saif Benjaafar, Zicheng Wang, and Xiaotang Yang. Human in the Loop Automation: Ride-Hailing with Remote (Tele-) Drivers. **Management Science**, 71.3, 2527-2543 (2025)

Link: <https://pubsonline.informs.org/doi/10.1287/mnsc.2022.01687>

*Saif Benjaafar, Zicheng Wang, and Xiaotang Yang. The Impact of Automation on Workers when Workers are Strategic: The Case of Ride-Hailing. **Manufacturing & Service Operations Management**, 27(5):1571-1586 (2025)

Link: <https://pubsonline.informs.org/doi/10.1287/msom.2023.0416>

Zhen Qiao, Zicheng Wang, Xiaotang Yang, Lun Yu. Stop Chasing Shortage: Optimal Batch Relocation in One-Way Sharing Systems. Working Paper, 2026

Link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7052658

HONORS AND AWARDS

University of Minnesota, Second Place in the Initiative on the Sharing Economy Student Competition.

2019

PROFESSIONAL ACTIVITIES

Ad hoc reviewer for Manufacturing & Service Operations Management, Naval Research Logistics, European Journal of Operational Research ($\times 2$), Probability in the Engineering and Informational Sciences, PLOS Computational Biology.

PRESENTATIONS

Learning to Price when Customers and Servers Abandon.

◦ Stochastic Networks Conference (Poster)

2026

◦ INFORMS Annual Meeting

2025

Inferring Relative Fitness of Tumor Cells from Reconstructed Genealogical Trees.

◦ Informs Applied Probability Society Conference

2025

◦ Research Institute for Interdisciplinary Sciences, Shanghai University of Finance and Economics

2025

Human in the Loop Automation: Ride-Hailing with Remote (Tele-) Drivers.

◦ Data-Driven Queueing Challenges Conference

2024

Clonal Diversity at Cancer Recurrence.

◦ Informs Applied Probability Society Conference

2023

Evolving Copy Number Gains Promote Tumor Expansion and Bolster Mutational Diversification.

◦ Research in Computational Molecular Biology Conference

2022

◦ Masonic Cancer Center Research Symposium (Poster)

2022

Autonomous Vehicles for Ride-Hailing.

◦ INFORMS International Conference

2025

◦ MSOM Conference

2024

◦ POMS Annual Conference

2023

◦ Stochastic Networks Conference (Poster)

2022

Large Deviations Analysis on Cancer Evolution.

◦ ISyE Student Seminar

2020

Splitting Algorithms for Rare Events of Semimartingale Reflecting Brownian Motions.

◦ POMS International Conference

2019

Two-sided Matching.

◦ POMS Annual Conference

2019

SKILLS

Programming/Software: Matlab, Python, R

Language: English (fluent), Chinese (native)

REFERENCES

Ruping Sun (Postdoc Advisor)

Assistant Professor

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